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### Ultrasound endoscope

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#### Abstract

An ultrasound endoscope includes: a balloon suction duct that is provided inside an insertion portion and that is where liquid inside a balloon is suctioned through, the balloon covering an ultrasound transducer positioned at a distal end of the insertion portion; a rigid portion that includes a balloon groove and an end surface positioned at an end portion of the rigid portion, the end portion being on a distal end side of the balloon groove, the end surface being a surface intersecting a longitudinal direction of the insertion portion, the end surface having an opening formed thereon, the opening communicating with the balloon suction duct; and a holder that is positioned on a distal end side of the rigid portion, the holder being configured to hold the ultrasound transducer.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION (1) This application is a continuation of International Application No. PCT/JP2020/005735, filed on Feb. 14, 2020, the entire contents of which are incorporated herein by reference.

### BACKGROUND

#### 1. Technical Field

(1) The present disclosure relates to ultrasound endoscopes.

#### 2. Related Art

(2) In the related art, liquid, such as degassed water, is injected into a balloon attached to an

ultrasound endoscope having an ultrasound transducer at a distal end of an insertion portion inserted into a subject, the balloon covering the outer periphery of the ultrasound transducer, and the balloon is brought into close contact with an observed region, such as the digestive tract (see, for example, Japanese Laid-open Patent Publication No. 2004-209044). After observation, the liquid in the balloon is suctioned from the inside of the balloon via a balloon suction duct that has been inserted inside the insertion portion.

## SUMMARY

(3) In some embodiments, an ultrasound endoscope includes: a balloon suction duct that is provided inside an insertion portion to be inserted into a subject and that is where liquid inside a balloon is suctioned through, the balloon covering an ultrasound transducer positioned at a distal end of the insertion portion; a rigid portion that includes a balloon groove to which a band of the balloon is attachable, and an end surface positioned at an end portion of the rigid portion, the end portion being on a distal end side of the balloon groove, the end surface being a surface intersecting a longitudinal direction of the insertion portion, the end surface having an opening formed thereon, the opening communicating with the balloon suction duct; and a holder that is positioned on a distal end side of the rigid portion, the holder being configured to hold the ultrasound transducer, the holder including a distal end portion having a first outer diameter, and a proximal end portion having a second outer diameter smaller than the first outer diameter, the proximal end portion being arranged adjacently to the opening.

(4) The above and other features, advantages and technical and industrial significance of this disclosure will be better understood by reading the following detailed description of presently preferred embodiments of the disclosure, when considered in connection with the accompanying drawings.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a schematic diagram illustrating a configuration of an ultrasound endoscope according to an embodiment;
- (2) FIG. 2 is a perspective view of a distal end portion of the ultrasound endoscope illustrated in FIG. 1;
- (3) FIG. 3 is a partial sectional view corresponding to a line A-A in FIG. 2;
- (4) FIG. 4 is a perspective view of the ultrasound endoscope illustrated in FIG. 2 as viewed from an opposite side opposite to a side where an opening is;
- (5) FIG. 5 is a view of the ultrasound endoscope illustrated in FIG. 2, the view being taken in a direction indicated by an arrow B;
- (6) FIG. 6 is a perspective view of a distal end portion of an ultrasound endoscope according to a first modified example;
- (7) FIG. 7 is a top view of a distal end portion of an ultrasound endoscope according to a second modified example;
- (8) FIG. 8 is an enlarged sectional view of bending pieces of a bending portion;
- (9) FIG. 9 is an enlarged view of a distal end portion in FIG. 8;
- (10) FIG. 10 is an enlarged sectional view of bending rubber;
- (11) FIG. 11 is an enlarged sectional view of bending rubber;
- (12) FIG. 12 is a sectional view of a distal end portion;
- (13) FIG. 13 is a sectional view corresponding to a line C-C in FIG. 12;
- (14) FIG. 14 is an enlarged perspective view of a screw receiver;
- (15) FIG. 15 is an enlarged view of a fixing screw;
- (16) FIG. 16 is a sectional view of a fixing screw; and

(17) FIG. 17 is a sectional view of a fixing screw.

## DETAILED DESCRIPTION

(18) Embodiments of an ultrasound endoscope according to the disclosure will be described hereinafter by reference to the drawings. The disclosure is not limited by these embodiments. The disclosure may be applied generally to ultrasound endoscopes that enable observation using balloons.

(19) Any elements that are the same or corresponding to each other are assigned with the same reference sign throughout the drawings, as appropriate. It also needs to be noted that the drawings are schematic, and relations between dimensions of each element therein and proportions between the elements therein may be different from the actual ones. The drawings may also include a portion that differs in its dimensional relations or proportions between the drawings.

### Embodiments

#### (20) Configuration of Ultrasound Endoscope

(21) FIG. 1 is a schematic diagram illustrating a configuration of an ultrasound endoscope according to an embodiment. An ultrasound endoscope **1** includes: an insertion portion **2** that has an imaging portion provided at a distal end of the insertion portion **2** and that is inserted into a subject; an operating portion **3** that is provided consecutively to the insertion portion **2** on a proximal end side of the insertion portion **2**; a universal cord **4** extending from a side portion of the operating portion **3**; and a connector **5** that is provided consecutively to the universal cord **4** and is connected to devices, such as: an observation device that controls the ultrasound endoscope **1**; and a light source device for supply of illumination light to the ultrasound endoscope **1**. In this specification, as illustrated in FIG. 1, a distal end side (away from the operating portion **3**) will be referred to as the “distal end side” and a proximal end side (toward the operating portion **3**) will be referred to as the “proximal end side”, the distal end side and proximal end side being along a longitudinal direction of the ultrasound endoscope **1**.

(22) The ultrasound endoscope **1** is an ultrasound endoscope for bronchi or digestive tracts.

(23) The insertion portion **2** has, in order from the distal end side, a distal end portion **2a**, a bending portion **2b** configured to be bendable according to operation of the operating portion **3**, and a flexible tube portion **2c** having flexibility. A proximal end of the flexible tube portion **2c** is provided consecutively to the operating portion **3** on the distal end side of the operating portion **3**.

(24) The operating portion **3** has a forceps insertion opening **3a** provided for insertion of, for example, a forceps needle, which is a treatment tool. A forceps insertion passage is provided inside the insertion portion **2** and the forceps insertion opening **3a** serves as an insertion opening of the forceps insertion passage. That is, the ultrasound endoscope **1** is an endoscope through which a treatment tool is able to be inserted.

#### (25) Configuration of Distal End Portion of Ultrasound Endoscope

(26) FIG. 2 is a perspective view of a distal end portion of the ultrasound endoscope illustrated in FIG. 1. The ultrasound endoscope **1** has an ultrasound transducer **21**, a rigid portion **22**, a holder **23**, an illumination lens **24**, and an objective lens **25**. FIG. 3 is a partial sectional view corresponding to a line A-A in FIG. 2. The ultrasound endoscope **1** includes a balloon suction duct **26**. FIG. 3 illustrates a balloon **6** that has been attached to the distal end of the insertion portion **2** to cover the ultrasound transducer **21**.

(27) The ultrasound transducer **21** is a convex ultrasound transducer and has plural piezoelectric elements regularly arrayed to form a convex circular arc. The ultrasound transducer **21** may be not necessarily a convex ultrasound transducer and may be a linear ultrasound transducer instead. The ultrasound transducer **21** converts a pulse signal input from an ultrasound observation device into ultrasound pulses and transmits the ultrasound pulses into a subject. Furthermore, the ultrasound transducer **21** converts ultrasound echoes reflected in the subject into an electric echo signal and outputs the electric echo signal to the ultrasound observation device.

(28) The rigid portion **22** has a balloon groove **221**, a first surface **222**, and a stepped portion **223**. A

band **61** of the balloon **6** is attachable to the balloon groove **221**. The first surface **222** is a surface that is positioned on the distal end side of the balloon groove **221** and that is along a direction intersecting a longitudinal direction of the insertion portion **2**, and has an opening **222a** formed thereon, the opening **222a** communicating with the balloon suction duct **26** along a direction in which the balloon suction duct **26** extends. A plane in which the opening **222a** is open is along a direction intersecting the longitudinal direction of the insertion portion **2**, as illustrated in FIG. **3**. Furthermore, the first surface **222** and an end portion of the opening **222a** are approximately orthogonal to each other. This is because a brush that has jutted out from the opening **222a** may be damaged if an angle formed by the first surface **222** and the end portion of the opening **222a** is an acute angle.

(29) FIG. **4** is a perspective view of the ultrasound endoscope illustrated in FIG. **2** as viewed from an opposite side opposite to a side where an opening is. As illustrated in FIG. **4**, the rigid portion **22** has an opening **224** formed therein. The opening **224** is positioned on the distal end side of the balloon groove **221** and communicates with a balloon water feeding duct through which liquid is fed into the balloon **6** covering the ultrasound transducer **21**. The opening **224** may be formed such that part of the opening **224** is in the balloon groove **221**, as long as the opening **224** is at a position where the opening **224** is completely covered by the balloon **6** when the balloon **6** has been attached.

(30) The holder **23** is positioned on the distal end side of the rigid portion **22** and holds the ultrasound transducer **21**. FIG. **5** is a view of the ultrasound endoscope illustrated in FIG. **2**, the view being taken in a direction indicated by an arrow B. As illustrated in FIG. **5**, the holder **23** has a second surface **231** along the longitudinal direction of the insertion portion **2**, and a third surface **232** continuous with the second surface **231** and along a direction intersecting the longitudinal direction of the insertion portion **2**. The first surface **222** and the third surface **232** are approximately orthogonal to each other. Being approximately orthogonal herein means that an angle formed between the first surface **222** and the third surface **232** is, for example, 60° or larger, and this angle is more preferably 80° or larger. A space **233** has been formed between the balloon **6**, the first surface **222**, and the third surface **232**. This space **233** is formed on the distal end side of the opening **222a** and extends to the opening **222a** in a direction orthogonal to the longitudinal direction of the insertion portion **2**. Furthermore, the space **233** is continuous along a direction (a direction orthogonal to the plane of paper of FIG. **3**) orthogonal to both of the longitudinal direction and the direction in which the balloon suction duct **26** extends.

(31) The illumination lens **24** irradiates the inside of the subject with illumination light emitted by the light source device via a light guide. The illumination lens **24** has, formed on its surface, a chamfered portion that has been slightly chamfered. The illumination lens **24** is positioned at the stepped portion **223** that is a depressed portion formed on a sloped surface **225**. In other words, the illumination lens **24** is positioned at the stepped portion **223** that has been stepped lower than the objective lens **25** formed on the sloped surface **225**. As a result, light output from the chamfered portion of the illumination lens **24** is prevented from: being incident on the objective lens **25**; and thereby causing flare. Therefore, there is no need for application of a black adhesive, for example, to the chamfered portion of the illumination lens **24** to cover the chamfered portion for prevention of flare, and workability will thus be improved and the field of view will not be hindered by such an adhesive. Furthermore, the stepped portion **223** is a depressed portion that is open at the distal end side. As a result, liquid that has adhered to the illumination lens **24**, for example, will flow out from the side that is open without staying in the stepped portion **223** and this prevents scattering of the illumination light due to the liquid.

(32) The objective lens **25** has been formed on the sloped surface **225**. The objective lens **25** condenses light reflected inside the subject and forms an image on an imaging plane of an imaging element. An image signal generated in the imaging element is transmitted to an endoscope observation device via a signal cable.

- (33) The balloon suction duct **26** is provided inside the insertion portion **2** inserted into the subject. Liquid inside the balloon **6** covering the ultrasound transducer **21** positioned at the distal end of the insertion portion **2** is suctioned through this balloon suction duct **26**.
- (34) The embodiment described above enables a distal end of the brush inserted from the proximal end side to jut out from the opening **222a** because the opening **222a** is formed along the direction in which the balloon suction duct **26** extends. Therefore, workability for washing the ultrasound endoscope **1** is improved because the brush inserted into the balloon suction duct **26** from the proximal end side is able to be jugged out from the opening **222a** and the inside of the balloon suction duct **26** is thereby able to be washed.
- (35) Furthermore, because the space **233** has been formed in this embodiment, the balloon **6** is prevented from being stuck to the opening **222a** and thus from failing to be deflated when liquid inside the balloon **6** is suctioned through the balloon suction duct **26**.
- (36) Furthermore, in this embodiment, the space **233** is continuous along the direction (the direction orthogonal to the plane of paper of FIG. 3) orthogonal to both of the longitudinal direction and the direction in which the balloon suction duct **26** extends. As a result, liquid in the balloon **6** flows into the opening **222a** from both of the directions along which the space **233** is continuous and the balloon **6** is thus prevented from sticking to the opening **222a**. The first surface **222** and the third surface **232** are preferably approximately orthogonal to each other for liquid in the balloon **6** to more easily flow into the opening **222a**.
- (37) Furthermore, in this embodiment, at least part of the first surface **222** is preferably positioned outward relative to the holder **23** as viewed from the distal end side of the insertion portion **2**. Positioning part of the first surface **222** more outward than the holder **23** forms a level difference between the first surface **222** and the holder **23** and thus prevents the balloon **6** from sticking to the opening **222a**.
- (38) Furthermore, in this embodiment, at least part of the opening **222a** is preferably positioned outward relative to the holder **23** as viewed from the distal end side of the insertion portion **2**. Positioning part of the opening **222a** more outward than the holder **23** prevents: the brush, which has jugged out from the opening **222a**, from bumping against the holder **23**; and movement of the brush from being hindered, and workability for washing is thus improved. Similarly, in this embodiment, the brush, which has jugged out from the opening **222a**, is prevented from bumping against the holder **23**, movement of the brush is thus prevented from being hindered, and workability for washing is thus improved, because the opening **222a** is open in the plane along the direction intersecting the longitudinal direction (the left-right direction in FIG. 3) of the insertion portion **2**.

#### First Modified Example

(39) FIG. 6 is a perspective view of a distal end portion of an ultrasound endoscope according to a first modified example. As illustrated in FIG. 6, an opening **222Aa** communicating with a balloon suction duct and an opening **222Ab** communicating with a balloon water feeding duct have been formed on a first surface **222A** of a rigid portion **22A**. The opening **222Aa** of the balloon suction duct and the opening **222Ab** of the balloon water feeding duct may be provided on one side as illustrated. This first modified example enables washing by allowing a brush to be jugged out from the opening **222Aa** or the opening **222Ab** for both the balloon suction duct and the balloon water feeding duct.

#### Second Modified Example

(40) FIG. 7 is a top view of a distal end portion of an ultrasound endoscope according to a second modified example. FIG. 7 illustrates, with a broken line, a cross section of a balloon **6** that has been attached to a distal end of an insertion portion **2**. As illustrated in FIG. 7, a first surface **222B** has been formed on both sides of a rigid portion **22B**. An opening communicating with a balloon suction duct has been formed on the first surface **222B**, which is on one side, and an opening communicating with a balloon water feeding duct has been formed on the first surface **222B**, which

is on the other side. Furthermore, a third surface **232B** and a space **233B** have been formed on both sides of a holder **23B**. This second modified example enables washing by allowing a brush to be juttied out from the opening for both the balloon suction duct and the balloon water feeding duct. (41) An example where the rigid portion **22** and the holder **23** are separately bodied has been described above with respect to the embodiment, but the rigid portion **22** and the holder **23** may be formed integrally with each other.

(42) Furthermore, an example where the balloon suction duct and the balloon water feeding duct are provided separately has been described above with respect to the embodiment, but one duct may have functions of both the balloon suction duct and the balloon water feeding duct.

(43) Configuration of Bending Portion of Ultrasound Endoscope

(44) FIG. **8** is an enlarged sectional view of bending pieces of a bending portion. As illustrated in FIG. **8**, the bending portion **2b** has a first piece **101** positioned at a distal end of the bending portion **2b**, an operating wire **102** that transmits operation received by the operating portion **3** to the bending portion **2b**, and a rivet **103** that bendably connects bending pieces together.

(45) FIG. **9** is an enlarged view of a distal end portion in FIG. **8**. As illustrated in FIG. **9**, the bending portion **2b** has a blade **104** covering the outer periphery of the bending portion **2b**, and a blade end fixing portion **105** that fixes a distal end of the blade **104** to the first piece **101**.

Furthermore, the first piece **101** has a small diameter portion **101a** having a small diameter than a distal end portion of the first piece **101**, the distal end portion having the largest diameter. The blade end fixing portion **105** is positioned at the small diameter portion **101a**. As a result, the bending portion **2b** is able to be reduced in diameter and strain on the patient at the time of insertion is able to be reduced, as compared to a case where the blade end fixing portion **105** is positioned at the distal end portion of the first piece **101**, the distal end portion having the largest diameter.

(46) FIG. **10** is an enlarged sectional view of bending rubber. Although not illustrated in FIG. **9**, bending rubber **106** is attached to the outer periphery of the blade **104** and keeps the inside of the insertion portion **2** watertight. As illustrated in FIG. **10**, thicknesses **t1**, **t2**, and **t3** of the bending rubber **106** have a relation,  $t1 < t2 < t3$ . A portion of the bending rubber **106**, the portion having the thickness **t1**, corresponds to the distal end portion of the first piece **101**, the distal end portion having the largest diameter. A portion of the bending rubber **106**, the portion having the thickness **t2**, corresponds to the small diameter portion **101a** and a portion on the proximal end side of the small diameter portion **101a**. A portion of the bending rubber **106**, the portion having the thickness **t3**, corresponds to a portion that is at the proximal end side and that is uniform in thickness.

(47) The bending portion **2b** is preferably bent uniformly from the proximal end side to the distal end upon bending operation. However, in a case where bending rubber is uniform in thickness, the proximal end side tends to be bent more. Therefore, the bending rubber **106** is configured such that the bending portion **2b** is bent uniformly from the proximal end side to the distal end upon bending operation, by adjustment of the thickness of the bending rubber **106**.

(48) FIG. **11** is an enlarged sectional view of bending rubber. As illustrated in FIG. **11**, in this modified example, thicknesses **t11**, **t12**, **t13**, and **t14** of bending rubber **106A** have relations,  $t11 < t12$  and  $t14 < t13$ . A portion of the bending rubber **106A**, the portion having the thickness **t14**, corresponds to a proximal end portion of the blade **104**, the proximal end portion being a portion to be fixed to the first piece **101**. The proximal end side of the bending rubber **106A** may thus be made thin and the bending portion **2b** may thereby be reduced in diameter.

(49) Internal Configuration of Distal End Portion of Ultrasound Endoscope

(50) FIG. **12** is a sectional view of a distal end portion. As illustrated in FIG. **12**, the holder **23** has a groove **234** formed on the proximal end side of the balloon groove **221** and a screw receiver **27** has been fitted in this groove **234**. Positioning the screw receiver **27** on the proximal end side of the balloon groove **221** enables reduction in rigid length (length of a portion that does not have flexibility and is rigid) at the distal end of the insertion portion **2** and reduction in strain on the

patient.

(51) FIG. 13 is a sectional view corresponding to a line C-C in FIG. 12. As illustrated in FIG. 13, screwing fixing screws 28 into the screw receiver 27 fixes the holder 23 to the rigid portion 22. Furthermore, the fixing screws 28 are sealed by fillers 29.

(52) FIG. 14 is an enlarged perspective view of a screw receiver. As illustrated in FIG. 14, the screw receiver 27 has, formed therein, female screws portions 27a where the fixing screws 28 are screwed into, and screw receiving portions 27b that come into contact with the fixing screws 28. The screw receiver 27 is made of, for example, metal or alloy.

(53) FIG. 15 is an enlarged view of a fixing screw. As illustrated in FIG. 15, each of the fixing screws 28 has, formed therein, a male screw portion 28a that is screwed into the female screw portion 27a and an abutting surface 28b that comes into contact with the screw receiving portion 27b. The male screw portion 28a and the abutting surface 28b of each of the fixing screws 28 have been formed integrally with each other. The fixing screws 28 are made of, for example, metal or alloy.

(54) Screwing the fixing screw into the screw receiver 27 causes the abutting surface 28b to come into contact with the screw receiving portion 27b. As a result, the screw receiver 27 and fixing screw 28, which are metallic, come into contact with each other and are thus firmly fastened to each other. Furthermore, a distal end of the fixing screw 28 is prevented from coming into contact with the holder 23 and causing distortion, for example.

(55) FIG. 16 is a sectional view of a fixing screw. As illustrated in FIG. 16, a fixing screw 28A has a screw 281A where a male screw portion 281Aa has been formed, and a sleeve 282A where an abutting surface 282Aa has been formed. The sleeve 282A for abutting may be fixed to the outer periphery of the screw 281A, as illustrated.

(56) FIG. 17 is a sectional view of a fixing screw. As illustrated in FIG. 17, a fixing screw 28B has a screw 281B, and a sleeve 282B where a male screw portion 282Ba and an abutting surface 282Bb have been formed. The sleeve 282B for screwing and abutting may be fixed to the outer periphery of the screw 281B, as illustrated.

(57) The disclosure enables an ultrasound endoscope to be provided, the ultrasound endoscope enabling: improvement in workability for washing of the balloon suction duct; and prevention of failure of balloon deflation.

(58) Additional advantages and modifications will readily occur to those skilled in the art. Therefore, the disclosure in its broader aspects is not limited to the specific details and representative embodiments shown and described herein. Accordingly, various modifications may be made without departing from the spirit or scope of the general inventive concept as defined by the appended claims and their equivalents.

## Claims

1. An ultrasound endoscope, comprising: a first duct provided inside an insertion portion; a rigid portion comprising: a balloon groove configured to accept a band of a balloon, and a first surface positioned distally relative to the balloon groove, the first surface intersecting a longitudinal direction of the insertion portion, the first surface having a first opening communicating with the first duct; and a casing positioned distally relative to the rigid portion, the casing configured to hold an ultrasound transducer; wherein a distal portion of the first duct overlaps a proximal portion of the casing in a longitudinal direction of the rigid portion, the rigid portion further comprising a second duct configured to feed liquid into the balloon covering the ultrasound transducer, and the second duct having a second opening on the first surface.
2. The ultrasound endoscope according to claim 1, wherein the casing includes an external surface, and at least part of the first surface extends radially outward relative to the external surface.
3. The ultrasound endoscope according to claim 1, wherein the casing includes an external surface,



- and at least part of the first opening is positioned radially outward relative to the external surface.
4. The ultrasound endoscope according to claim 1, wherein the casing includes an external surface, the external surface and the first surface together define a space, and wherein the first opening of the first duct is in fluid communication with the space.
  5. The ultrasound endoscope according to claim 4, wherein the external surface has a second surface extending along the longitudinal direction of the rigid portion and a third surface continuous with the second surface, the third surface extending proximally and radially inward from the second surface, and the space is formed between the first surface, and the third surface.
  6. The ultrasound endoscope according to claim 5, wherein an angle between the first surface and the third surface is at least 60°.
  7. The ultrasound endoscope according to claim 5, wherein the space is continuous along a direction orthogonal to both of the longitudinal direction of the rigid portion and a direction in which the first duct extends.
  8. The ultrasound endoscope according to claim 4, wherein an entirety of the space is offset radially from a central longitudinal axis of the insertion portion.
  9. The ultrasound endoscope according to claim 4, wherein the first surface and the external surface intersect to form the space.
  10. The ultrasound endoscope according to claim 1, wherein the first opening is open in a plane along a direction intersecting the longitudinal direction of the insertion portion.
  11. The ultrasound endoscope according to claim 1, further comprising a fourth surface radially offset from the first surface relative to the longitudinal direction of the insertion portion, wherein the first opening and the second duct having a second opening on the fourth surface.
  12. The ultrasound endoscope according to claim 1, further comprising the balloon, the balloon covering the ultrasound transducer positioned at a distal end of the insertion portion.
  13. The ultrasound endoscope according to claim 1, wherein the casing includes: a distal end portion having a first outer diameter; and a proximal end portion having a second outer diameter smaller than the first outer diameter, the proximal end portion arranged adjacently to the first opening.
  14. The ultrasound endoscope according to claim 1, wherein the first duct is configured to suction liquid inside the balloon.
  15. The ultrasound endoscope according to claim 1, wherein the rigid portion comprises a projection, the first surface is a distal end surface of the projection.
  16. The ultrasound endoscope according to claim 15, wherein an entirety of the projection is offset radially from a central longitudinal axis of the insertion portion.
  17. The ultrasound endoscope according to claim 1, wherein the first opening of the first duct is offset radially from the longitudinal axis of the insertion portion.
  18. The ultrasound endoscope according to claim 1, wherein the casing is fixed to the rigid portion.
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